

Patrick Tierney

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EXPERIENCE

Microsoft

Software Engineer II (2019 – 2025), **Software Engineer I** (2017 – 2019)

Infrastructure & Developer Systems

- Architected and led delivery of a testing platform used by 150+ engineers supporting Copilot Hub, standardizing automated testing
- Co-designed and implemented a telemetry platform for Microsoft Whiteboard using OpenTelemetry, making the service easier to diagnose in production while keeping developer overhead low
- Consolidated disparate AI tooling from across Microsoft teams into a unified, maintainable platform codebase
- Defined standards and requirements for an AI-driven static analysis system enforcing testing and compliance
- Enhanced a telemetry query DSL used by 1,000+ Edge engineers, improving analytics and diagnostic capabilities
- Drove migration of Whiteboard's layout engine from C# to TypeScript/React within its backing framework
- Built layout-engine tooling to position visual elements correctly on screen, supporting 80+ engineers as Whiteboard scaled to tens of millions of users
- Owned layout logic for on-screen placement of non-hierarchical graphical elements across 50+ internal applications

Reliability & Observability

- Established on-call and incident response practices to address the root cause behind recurring issues, reducing incident resolution time by 80% and engineering support workload by 70%
- Maintained 99.9% availability for Microsoft Whiteboard supporting school systems globally during COVID-19
- Designed support and escalation model for sovereign government cloud environments (GCC, GCCH, DoD)
- Built tooling and operational runbooks enabling customer support to resolve issues without engineer escalation
- Developed automation that generated dashboards from telemetry data for new feature rollouts

Production Services & Distributed Systems

- Developed a public REST API surface and services architecture enabling Microsoft Whiteboard CRUD workflows
- Deployed and maintained authentication and authorization services for Whiteboard and Copilot Hub
- Maintained and Optimized the Edge Telemetry Data Processing Pipeline
- Built and managed CI/CD for Whiteboard and Copilot Hub, automating builds, testing, and validation

TECHNICAL SKILLS

Languages & Frameworks: TypeScript, JavaScript, Node.js, C#, .NET, SQL, React, Python, C++, R

Infrastructure & Automation: Azure services, Azure DevOps, YAML-based CI/CD, PowerShell

Platform, Reliability & Security: OpenTelemetry, Distributed Tracing & Observability, CI/CD systems, Incident Response & On-Call Operations, OAuth/JWT authentication, Jest, Playwright, Power BI

EDUCATION

Harvey Mudd College

Bachelor's Degree, Computer Science and Mathematics; Concentration in Economics

Claremont, CA

May 2016